

### Hamiltonian Path

A Hamiltonian path is a path in a connected graph that includes **every vertex** in the graph once only and it starts and finishes at different vertices.

Consider the graph on the right.

The path ABCDEGF shown on the graph is a Hamiltonian path as it includes every vertex once, begins and ends at different vertices.

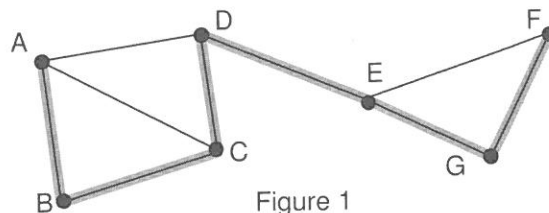


Figure 1

The Hamiltonian path shown in the graph is not the only Hamiltonian path, there are many others for example BCADEGF, GFEDABC, etc.

Note: If bridge DE was removed the graph will be a disconnected graph and will no longer have a Hamiltonian path. Therefore Hamiltonian paths are only possible for connected graphs.

### Hamiltonian Cycle

If a Hamiltonian path is closed, that is it begins and ends with the same vertex, then the graph is said to be **Hamiltonian** and the closed path (or cycle) is said to be a Hamiltonian cycle.

Thus we can say that a **Hamiltonian cycle** is a cycle that includes every vertex in a graph once only and it begins and ends at the same vertex.

Consider the graph on the right.

The path ABEDCA is a closed Hamiltonian path beginning and ending at vertex A. Hence the path ABEDCA is a Hamiltonian cycle and the graph is said to be Hamiltonian.

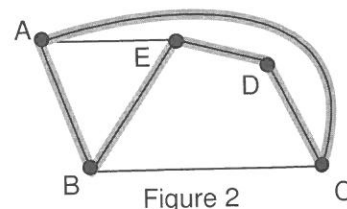


Figure 2

The Hamiltonian cycle shown in the graph is not the only Hamiltonian cycle there are many others for example EDCBAE, CBAEDC, etc.

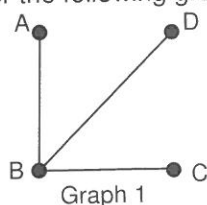
Note that if a graph has a Hamiltonian cycle it also has a Hamiltonian path. For example ABEDCA is a Hamiltonian cycle and ABEDC is a Hamiltonian path.

### Semi-Hamiltonian graph

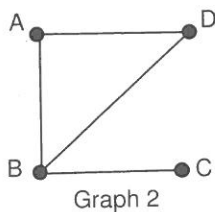
A connected graph that contains a Hamiltonian path but not a Hamiltonian cycle is said to be a semi-Hamiltonian graph.

Figure 1 at the top of the page is a semi-Hamiltonian graph because it contains a Hamiltonian path but no Hamiltonian cycle.

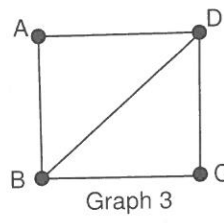
Consider the following graphs:



Graph 1



Graph 2



Graph 3

**Graph 1** does not have a Hamiltonian path as it is not possible to visit every vertex once only, and therefore cannot have a Hamiltonian cycle.

**Graph 2** has a Hamiltonian path DABC (not the only one) but has no Hamiltonian cycle.

**Graph 3** has a Hamiltonian path ABCD and it also has a Hamiltonian cycle ABCDA. Hence graph 3 is Hamiltonian.

### Properties of Hamiltonian Graphs

- Hamiltonian graphs are connected graphs
- any Hamiltonian cycle can be converted into a Hamiltonian path by removing one of its edges
- a Hamiltonian path can be converted to a Hamiltonian cycle if, and only if, the endpoints of the Hamiltonian path are adjacent.
- all complete graphs, that is  $K_n$  are Hamiltonian
- complete bipartite graphs which have the same number of vertices in each group are Hamiltonian

### Determining if a Graph is Hamiltonian or semi-Hamiltonian

There is no general test for determining the presence of Hamiltonian paths and cycles. To determine if a graph is Hamiltonian or semi-Hamiltonian we need to check every vertex.

However, for a graph to be Hamiltonian or semi-Hamiltonian it must be a connected graph and it can have no more than two vertices of degree 1.

### Eulerian and Hamiltonian Graphs

An **Eulerian trail** is a trail that passes along every **edge** exactly once.

If an Eulerian trail ends at the starting vertex then it is said to be an **Eulerian circuit**.

An **Eulerian graph** is a connected graph which is traversable, starting and finishing at the same vertex. Eulerian graphs must have no odd vertices.

If an Eulerian trail does not end at the starting vertex then it is said to be a **semi-Eulerian trail**.

A **semi-Eulerian graph** is a connected graph which is traversable, starting and finishing at different vertices. Semi-Eulerian graphs must have exactly two odd vertices.

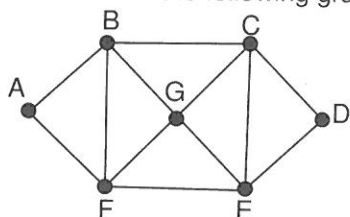
A **Hamiltonian path** is a path that passes through each **vertex** exactly once.

If a Hamiltonian path ends at the starting vertex it is said to be a **Hamiltonian cycle**.

A **Hamiltonian graph** is a connected graph which has a Hamiltonian cycle.

A **semi-Hamiltonian graph** is a connected graph which has a Hamiltonian path but not a Hamiltonian cycle.

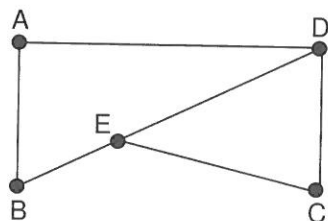
Examine each of the following graphs and their classifications.



The graph is Eulerian as there are no odd vertices.

The graph has Euler circuits e.g. ABCDEFGCEGBFA.

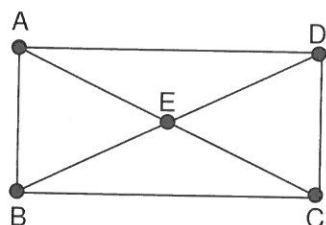
The graph is Hamiltonian as it has Hamiltonian cycles, for example one such Hamiltonian cycle is ABCDEGFA. The graph also has Hamiltonian paths for example ABCDEGF.



The graph is semi-Eulerian as it has exactly two odd vertices.

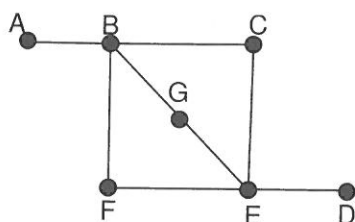
The graph does not have any Eulerian circuits but has semi-Eulerian trails, for example DABEDCE.

The graph is Hamiltonian as it has Hamiltonian cycles, one such cycle is ECDABE. It also has Hamiltonian paths.



The graph is not Eulerian or semi-Eulerian as it has more than two odd vertices.

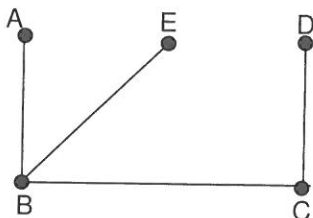
The graph is Hamiltonian as it has Hamiltonian cycles one such cycle is EABCDE. It also has Hamiltonian paths, one such Hamiltonian path is EABCD.



The graph is semi-Eulerian as it has exactly two odd vertices.

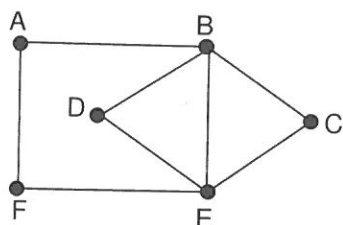
The graph does not have any Eulerian circuits but has semi-Eulerian trails, for example ABFEGBCED.

The graph is not Hamiltonian or semi-Hamiltonian as it has neither Hamiltonian cycles nor Hamiltonian paths.



The graph is not Eulerian or semi-Eulerian as it has more than two odd vertices.

The graph is not Hamiltonian or semi-Hamiltonian as it has neither Hamiltonian cycles nor Hamiltonian paths.



The graph is Eulerian as there are no odd vertices.

The graph has Euler circuits e.g. AFECBEDBA.

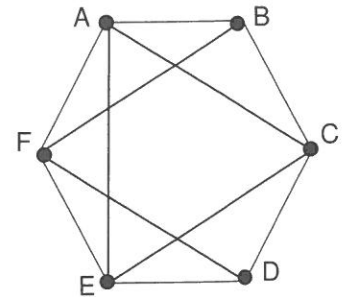
The graph is semi-Hamiltonian as it has Hamiltonian paths but no Hamiltonian cycles. One Hamiltonian path is CEDBAF.

The graphs considered on this page tell us that the existence of an Euler circuit or semi-Euler trail in a graph tells us **nothing** about the existence of a Hamiltonian path or Hamiltonian cycle in the graph.

## EXERCISE 8C

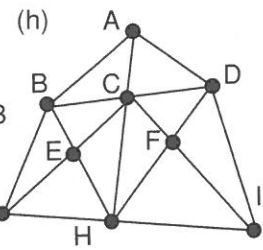
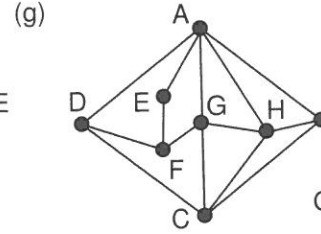
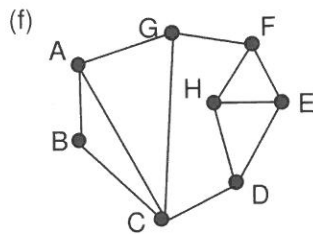
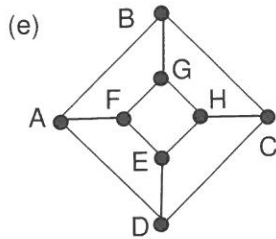
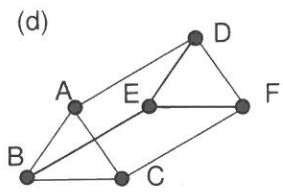
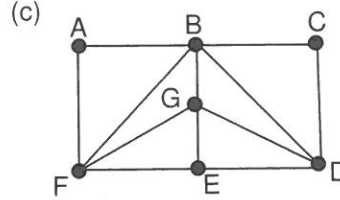
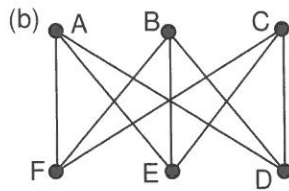
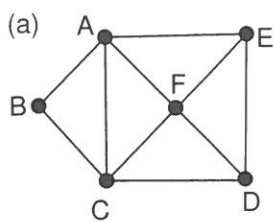
1. Explain what it means for a graph to have a Hamiltonian cycle.

2. For the given graph on the right, determine which of the following paths are:  
 (i) Hamiltonian paths (iii) neither Hamiltonian paths or Hamiltonian cycles  
 (ii) Hamiltonian cycles

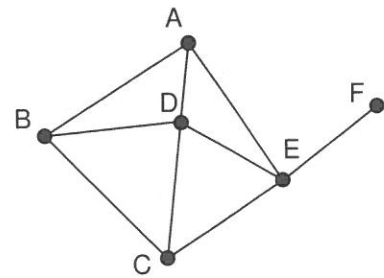


- (a) ABCEDF (e) CABFED  
 (b) FBDCEA (f) CDEAFBC  
 (c) BCDEAFECABF (g) AEDCBFEA  
 (d) DCEFBA (h) AFEDCBFA

3. List a Hamiltonian cycle for each of the following graphs.

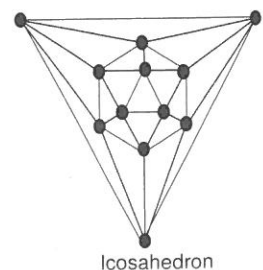
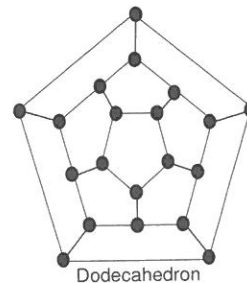
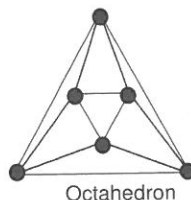
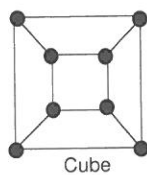
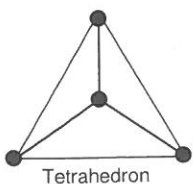


4. Consider the graph on the right.  
 (a) Does a Hamiltonian cycle exist on the graph?  
 If yes, list the vertices of the Hamiltonian cycle.  
 If no, justify.



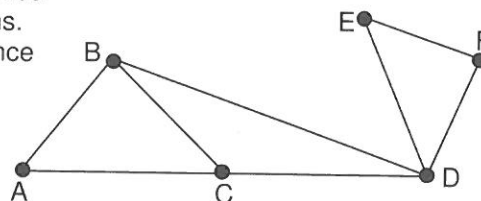
- (b) Does a Hamiltonian path exist on the graph?  
 If yes, list the vertices of the Hamiltonian path.  
 If no, justify.

5. The planar graphs of the five platonic solids are given below. All graphs of the platonic solids are Hamiltonian, on each of the Hamiltonian graphs show a Hamiltonian cycle.



6. Which of the platonic solids is Eulerian? Justify your response.
7. A complete graph,  $K_n$  where  $n$  is the number of vertices, is one in which there is an edge connecting every vertex to every other vertex.  
For what values of  $n$  does a complete graph have:  
(a) a Hamiltonian cycle? (b) an Eulerian circuit?

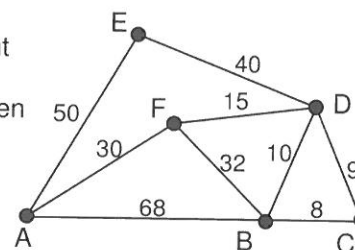
8. The graph on the right shows the location of six towns labelled A, B, C, D, E and F, and the roads that connect these towns.  
(a) Can a salesperson start from town A visit each town once and return to town A? Justify your response.



- (b) The given graph is semi-Hamiltonian.  
What does this mean?

- (c) Is the graph Eulerian, semi-Eulerian or neither? Justify your response.

9. The map of six towns is represented by the graph shown on the right. The vertices of the graph represent the towns and the edges represent the roads connecting these towns. The numbers on the edges indicate the distance, in kilometres, between these towns.



- (a) What type of graph represents the map of the towns?

A courier stationed at A has to make deliveries to each of the towns and return to A.

- (b) Find the length, in kilometres, of a delivery run that follows a Hamiltonian cycle from town A.  
List the vertices of the Hamiltonian cycle starting at A.

Roadworks restrict traffic along road BD. Traffic can only travel one-way, from D through to B.

- (c) What type of graph represents the map of the towns with this restriction in place?

The shortest possible circuit from town A for this courier run is not Hamiltonian.

- (d) List the vertices of this circuit and state its length.

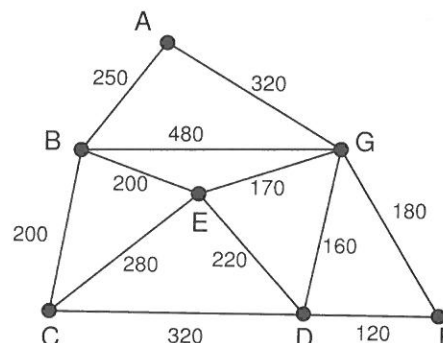
- (e) Explain why this shortest circuit is not Hamiltonian.

10. Clive, a tourist, has identified seven places of interest he would like to visit during his stay in Rome. On the network diagram the vertices represent the places of interest and the edges the walking paths between these places.

The numbers on the edges represent the lengths of the paths in metres.

Clive decides to go for a walk that will take him along each of the paths between the places of interest. He wishes to walk the minimum distance.

- (a) Where should Clive begin his walk?  
(b) What total distance will Clive walk?

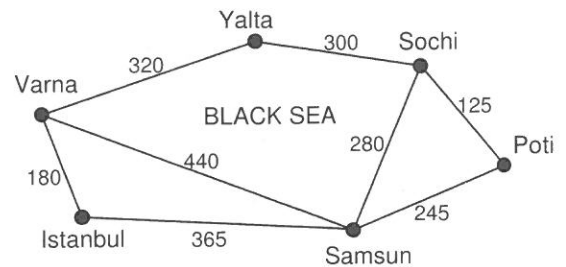


Clive's wife, Kim, is at site A and wished to visit each place of interest taking a route that can be described as the shortest Hamiltonian cycle.

- (c) Write down a route that Kim can take.

11. A complete bipartite graph is denoted by the notation  $K_{m,n}$  where  $m$  is the number of vertices in one group and  $n$  the number of vertices in the other group. For what values of  $m$  and  $n$  does the complete bipartite graph  $K_{m,n}$  have:
- (a) a Hamiltonian cycle? (b) an Eulerian circuit?
- (HINT: Investigate by drawing systematic complete bipartite graphs)

12. Varna, Yalta, Sochi, Poti, Samsun and Istanbul are six ports on the Black Sea. The network shows the sea routes and distances between these ports in nautical miles.
- (a) What is the shortest sea route distance, in kilometres, between Istanbul and Yalta? (1 nautical mile = 1.852 km)



- (b) How many different ways are there to sail from Istanbul to Yalta without passing through any port more than once? List these ways.

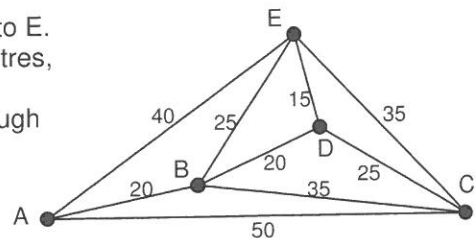
A cruise is planned to travel along each sea route once only in this network.

- (c) Where should the cruise start and finish?

A special cruise is planned to start and finish at the Russian tourist port of Sochi. This special cruise will exclude two of the sea routes between these ports but still travel along each of the remaining six sea routes and visit each of the other five ports once only.

- (d) How much shorter, to the nearest kilometre, will this cruise be compared to the cruise in (c)?

13. The graph shows a network of roads between five towns A to E. The numbers on the edges indicate the distances, in kilometres, between these towns. Pam, a mathematics teacher, followed an Eulerian trail through this network of roads.



- (a) Where did Pam start and where did she finish?
- (b) List a possible route that Pam took.
- (c) What distance did Pam cover?

Pam decides to travel from A to E following a Hamiltonian path.

- (d) What is the shortest distance that she can travel?

Recent rains caused a landslide and the road connecting D and E had to be closed.

- (e) The closure of which other road will allow Pam to follow a Eulerian circuit through this network? Justify your response.